

# Personalized AI assistants could either promote or hinder the emergence of collective intelligence

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## Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

## Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of ”tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly penalizing or incentivizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extended previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allowed each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

## Results

### Model

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target  $Y$  is modeled as a linear combination of  $M$  independent factors  $(x_1, x_2, \dots, x_m)$  and  $M + 1$  coefficients  $(\alpha_0, \alpha_1, \dots, \alpha_m)$ , where  $\alpha_0$  denotes the global offset (Fig. 1A). The  $M$  factors are randomly drawn from normal distributions with mean  $X_i$  and standard deviation  $\sigma_i$  in each generation.

Among a population of  $N$  individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual  $A$ 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (1)$$

where  $c_A$  represents individual  $A$ 's personal belief,  $e_A$  represents the factor that  $A$  observes (interest), and  $x_{e_A}$  represents the realized value of  $A$ 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let  $x_0$  always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor  $i \in [0, M]$ , AI knows the exact coefficient  $\alpha_i$  and has a fixed bias  $b_i$ , resulting in a prediction of  $\alpha_i x_i + b_i$ . In terms of human-AI interaction, we model two contrasting modes of AI: the "AI knows all" model and the "AI answers question" model. In the AI knows all model, AI acts as an omniscient system and gives an answer regardless of the individual's interest, accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (2)$$

In contrast, the AI answers question model acts as a chatbot and gives an answer only based on the individual's interest. Thus, given the individual's interest,  $e_A$ , AI predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (3)$$

In both models,  $\epsilon \sim N(0, \sigma_e^2)$  represents random error arising from the stochastic generative process of the AI model.

AI works as a personalized assistant for each individual, such that each individual receives assistance by incorporating the AI's prediction. However, even with the same opportunity to receive AI assistance, reliance on AI can vary substantially across individuals. To model this, we assume that the revised prediction is a weighted sum of the individual's own prediction and the AI's prediction by introducing AI belief

97  $\beta_A$  as a weighting coefficient. The revised prediction of individual  $A$  can be written  
 98 as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (4)$$

99 The revised predictions of individuals are aggregated by two different mechanisms,  
 100 depending on the mode of AI. In the AI knows all model, even if an individual can  
 101 observe only a single factor, their revised prediction contains information about all  
 102 factors through interaction with AI. Therefore, we use simple averaging of revised  
 103 predictions as the collective prediction. In contrast, in the AI answers question  
 104 model, each individual gives a prediction based only on a single factor even after  
 105 revising their prediction with AI assistance. Therefore, we use clustering aggregation  
 106 to obtain the collective prediction, which is performed by clustering individuals based  
 107 on interest and summing the average values of each cluster.

108 For each generation, each individual is rewarded with a payoff based on the given  
 109 incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and  
 110 “niche expert,” which were proposed by Wang et al. [13] and promote collective  
 111 intelligence when prediction is performed without AI. The feedback incentive gives  
 112 higher payoff to individuals who can push the collective prediction toward the true  
 113 value, regardless of their individual performance. Given individual  $A$ ’s revised pre-  
 114 diction  $Y_A^{revised}$  and the collective prediction  $\hat{Y}^{revised}$ , it can be expressed as:

$$\pi_A = Y_A^{revised} (Y - \hat{Y}^{revised}) \quad (5)$$

115 The niche expert incentive gives higher payoff to individuals who have both better  
 116 individual performance and a minority interest. Specifically, the niche expert payoff  
 117 is expressed as the product of the proportion of individuals who observe the same  
 118 factor as individual  $A$ ,  $\rho_{e_A}$ , and individual accuracy. Given individual  $A$ ’s revised  
 119 prediction  $Y_A^{revised}$ , it can be expressed as:

$$\pi_A = -\rho_{e_A} (Y_A^{revised} - Y)^2 \quad (6)$$

120 These two incentive structures were shown to achieve nearly perfect collective in-  
 121 telligence through social learning (except for the niche expert incentive with simple  
 122 averaging aggregation) by promoting individual expertise while maintaining collec-  
 123 tive diversity. Therefore, we use these incentive structures as a baseline to under-  
 124 stand how AI affects the emergence of collective intelligence. Moreover, we modify  
 125 the two baseline incentives in a later section to understand the consequences of direct  
 126 incentives and penalties for AI usage.

127 Throughout the simulation, individuals engage in social learning by imitating  
 128 other individuals who receive higher payoffs. We assume that prediction events  
 129 occur more frequently than social learning events within the population, such that  
 130 individual payoff can be computed as an expected payoff. The imitation process  
 131 is performed for at most one pair in each generation. By sampling two random  
 132 individuals,  $A$  and  $B$ , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (7)$$

where  $s$  denotes the strength of social learning. When imitation occurs,  $A$  imitates all three strategies of  $B$ : interest, belief, and AI belief.

To evaluate and analyze the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, human accuracy, interest diversity, and median AI belief. These metrics capture the accuracy of the revised collective prediction, its bias–variance decomposition, the predictive performance without AI, the diversity of individual interests, and median AI belief in the population (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

| Metric              | Definition                                                                              | Interpretation                                                                                    |
|---------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Collective accuracy | $1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$ | Accuracy of the revised collective prediction relative to the baseline prediction $\alpha_0$ .    |
| Collective bias     | $\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$                               | Squared systematic deviation of the revised collective prediction from the true value.            |
| Collective variance | $\text{Var}(Y - \hat{Y}^{\text{revised}})$                                              | Variability of the revised collective prediction error across prediction events.                  |
| Human accuracy      | $1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$                  | Accuracy of the human-only collective prediction relative to the baseline prediction $\alpha_0$ . |
| Interest diversity  | $\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$                                      | Effective number of interests represented in the population.                                      |
| Median AI belief    | $\text{median}_{A \in N}(\beta_A)$                                                      | Median value of AI belief $\beta_A$ in the population.                                            |

## A parsimonious model for RSV epidemics

We first began by asking whether a simple Susceptible-Infected-Recovered-Susceptible (SIRS) model can capture the observed RSV outbreak dynamics in Japan, including

the sudden change in outbreak seasonality. The SIRS model is the simplest dynamical model that allows for the possibility of immune waning and therefore represents one of the most parsimonious models for explaining outbreak dynamics of respiratory infections. Here, we extended the standard SIRS model such that we could simultaneously estimate periodic seasonal transmission rates and non-periodic changes in transmission due to NPI measures that were implemented to prevent COVID-19 (Materials and methods).

Accounting for flexible changes in seasonal transmission rates that deviate from standard sinusoidal shapes allowed the SIRS model to reproduce the observed dynamics across all five islands, including changes in seasonality that occurred during 2016–2017 as well as post-pandemic changes in outbreak patterns (Fig. 2A). In contrast to most seasonal respiratory pathogens, which exhibit an annual cycle in transmission pattern, we estimated semiannual peaks in transmission rates in four islands: Honshu, Shikoku, Kyushu, and Ryukyu (Fig. 2B). These semiannual peaks in transmission rates were explained by the quadratic effects of specific humidity: in line with [? ], we estimated that transmission would be maximized at a low and high mean specific humidity and minimized at intermediate mean specific humidity (Fig. 2C). Interestingly, we found that minimum RSV transmission occurred at a much higher mean specific humidity in Ryukyu island than in other three islands (Fig. 2C). We did not find semiannual transmission rate patterns or quadratic humidity-transmission relationship for Hokkaido island (Fig. 2B–C); instead, we found that transmission decreased at very low specific humidity ( $< 5$  g/kg). Differences in the ranges of observed mean specific humidity as well as the humidity-transmission relationship likely reflect heterogeneity in climate conditions. In particular, Honshu, Shikoku, and Kyushu islands are clustered around the main part of Japan, whereas Hokkaido and Ryukyu islands are located in the north and south from main islands, respectively (Supplementary Figure S2). Combining transmission rate estimates from all five islands still gave a quadratic humidity-transmission relationship for specific humidity between  $\approx 5$  g/kg and  $\approx 18$  g/kg, but the joint relationship poorly captured the humidity-transmission relationship in the Ryukyu island (Supplementary Figure S3).

We also found similar quadratic relationships between the estimated transmission rates and mean temperature (Supplementary Figure S4). Performing bivariate quadratic regressions against both humidity and temperature showed significant, positive effects of the quadratic humidity term in Honshu ( $p = 0.01$ ) and Ryukyu ( $p < 0.01$ ) islands (Supplementary Table S1). We found positive effects of the quadratic humidity term in Kyushu island but this effect was not significant ( $p = 0.27$ ; Supplementary Table S1). We found negative effects of the quadratic humidity term in Shikoku island but this effect was almost close to zero and not significant ( $p = 0.67$ ; Supplementary Table S1). However, we note that mean temperature and mean specific humidity are almost perfectly correlated, with correlation coefficients  $> 0.96$  in all islands (Supplementary Figure S5), and therefore the bivariate regression cannot tease apart the effects of humidity and temperature separately.

187 We also compared R squared values from two separate quadratic regression mod-  
 188 els to test their ability to explain the variation in the estimated transmission rates  
 189 using mean specific humidity and mean temperature as separate covariates (Supple-  
 190 mentary Table S2). The humidity model outperformed the temperature model for  
 191 Honshu (0.39 vs 0.29) and Ryukyu (0.59 vs 0.47) islands, whereas the temperature  
 192 model outperformed the humidity model for Hokkaido (0.44 vs 0.52) and Shikoku  
 193 (0.63 vs 0.72) islands. Both models had nearly identical R squared values for Kyushu  
 194 islands (0.77 vs 0.77).

195 Across all five islands, we estimated large reductions in transmission during 2020  
 196 (Fig. 2D); however, there was large heterogeneity in the overall shape of the esti-  
 197 mated NPI effects as well as the degree of transmission reduction. The reduction in  
 198 transmission rates caused an increase in the susceptible pool (Fig. 2E), which allowed  
 199 a large outbreak when NPIs were lifted (Fig. 2A). This observation is also consistent  
 200 with [?] who predicted that an accumulation of the susceptible host population  
 201 during the NPI period will eventually lead to a large outbreak.

## 202 Mechanisms for sudden changes in seasonality

203 Since the mechanistic SIRS model was able to accurately capture the observed transi-  
 204 tion in seasonality, we posit that it captures the relevant mechanisms for driving this  
 205 pattern. Thus, we should be able to use the inferred dynamics to further tease apart  
 206 the mechanisms underlying sudden changes in seasonality of the RSV outbreaks.  
 207 To do so, we first began by evaluating the changes in the proportion of susceptible  
 208 and infected individuals at the beginning of the season between 2013 and 2019. We  
 209 focused on three islands where the sudden transition in RSV outbreak seasonality  
 210 from winter to fall was observed: Honshu, Shikoku, and Kyushu islands.

211 Across three main islands, we found a consistent increase in the proportion of  
 212 susceptible and infected individuals at the beginning of each season between 2013  
 213 and 2019 (Fig. 3A). These changes also corresponded with a decrease in center of  
 214 gravity (Fig. 3A). A more detailed comparison of epidemic trajectories illustrated  
 215 that an increase in the susceptible pool at the beginning of the season can drive  
 216 a sudden shift in seasonality (Fig. 3B). The semiannual pattern in transmission  
 217 (i.e., two peaks in transmission rates within a year) allows this transition, where a  
 218 faster epidemic growth (from higher susceptible pool) drives a faster depletion of  
 219 susceptibles and causes the epidemic to transition from a later peak to an earlier  
 220 peak.

221 To understand why we observe a bigger shift in seasonal outbreak patterns in  
 222 Kyushu island than in Honshu and Shikoku islands, we characterized how differences  
 223 in the seasonal transmission patterns affects the difference in peak epidemic timing  
 224 associated with an increase in population-level susceptibility (Fig. 3C–G). As a refer-  
 225 ence, we began with smoothed transmission rate estimates for Honshu (Fig. 3C) and  
 226 Shikoku (Fig. 3F) islands and explore intermediate transmission patterns that inter-  
 227 polate two transmission patterns. Specifically, the transmission pattern in Kyushu

exhibited a bigger amplitude and a wider trough between two transmission peaks. These differences were explored by varying the amplitude of the seasonal transmission rate ( $x$ -axis on Fig. 3G) and the degree of interpolation ( $y$ -axis on Fig. 3G), where the interpolation coefficients of 0 and 1 correspond to the seasonality in Honshu and Kyushu islands, respectively. Simulating the SIRS model using interpolated seasonal transmission rates showed that a large seasonal amplitude and wider trough cause larger changes in the timing of epidemic peak (Fig. 3G).

We did not observe a shift in RSV seasonality in Ryukyu island (Fig. 1E) despite the inferred quadratic humidity-transmission relationship (Fig. 2B). In Supplementary Materials, we show that there was limited variation in population-level susceptibility between 2014–2019 in Ryukyu island, explaining the lack of change in RSV seasonality (Supplementary Figure S6A). Likewise, we estimate a near constant susceptibility for 2013–2016 and 2019 for Hokkaido island and a small reduction in susceptibility during 2017 and 2018 (Supplementary Figure S7A).

Based on these observations, we tested whether an increase in transmission can also explain a sudden transition in RSV seasonality in Honshu, Shikoku, and Kyushu islands (Supplementary Figure S8). This was done by estimating a separate seasonal transmission term before and after the change in RSV outbreak seasonality. We found that an increase in transmission was also able to capture the shift in RSV outbreak seasonality (Supplementary Figure S8A,B). Under this model, we estimated a moderate increase in mean transmission across three islands: 23% (95% CI: 19%–27%) for Honshu island, 16% (95% CI: 13%–18%) for Kyushu island, and 14% (95% CI: 7%–20%). The estimated shape of seasonal transmission remained largely unchanged (Supplementary Figure S8B).

## Mechanisms for an increase in RSV transmission

The simple SIRS model predicted an increase in the proportion of susceptible individuals from 2013 to 2019 across Honshu, Shikoku, and Kyushu islands. As the SIRS model relies on simplifying assumptions about immunity against RSV infections, the estimated increase in the proportion of susceptible individuals likely reflects an effective increase in population-level susceptibility, rather than a strict increase in the raw number of susceptible individuals. This population-level susceptibility can be thought of as an average of how susceptible each person is to RSV infections across the population. Many factors can contribute to this population-level susceptibility, including immunological factors, such as the decreased probability of acquiring RSV following a primary RSV infection [? ], and behavioral factors, such as increased exposure to RSV from increased contact rates. Alternatively, we found that an increase in transmission can also explain the sudden transition in RSV seasonality. Both models indicate an increase in the overall rate of RSV transmission ( $\beta SI/N$ ) from 2013 to 2019. So what mechanisms caused the RSV transmission to increase over time in Japan?

Previously, [? ] hypothesized that changes in climate and an increase in inbound



overseas travelers may be both responsible for this shift in seasonality. While an increase in overseas travelers may also contribute to the increase in contact rates and therefore RSV transmission, it is likely to have weak effects given that the mean age of infection for RSV is typically very young. For example, a local surveillance effort in Kyoto reported that RSV infections were most frequently detected among < 6-year old (15.4% detection rate) compared to older age groups: 6–17 years (2.3%), 18–64 years (0.8%) and  $\geq 65$  years (0.6%) [? ].

Alternatively, we hypothesized that the Comprehensive Support System for Children and Childcare, which was enacted in 2012 and launched in 2015, brought more children into the daycare system and thus increased the probability of exposure to RSV, leading to an increase in RSV transmission. This hypothesis builds on the work of [? ] who previously suggested that increase in childcare attendance may have contributed to a large change in the seasonality of Kawasaki disease in Japan in the mid-2010s. An expansion of childcare facilities would increase contact rates among children, which in turn would increase the probability of exposure and therefore the population-level susceptibility against RSV infections. To quantify the potential impact of this program, we compared the number of children attending childcare facilities in Japan since 2013 and compared them with our estimates of susceptible proportion (Fig. 4). Overall, we found consistent patterns of increase in childcare attendance and strong correlations with the estimated susceptible proportion: 0.977 (95% CI: 0.847–0.997) in Honshu island, 0.943 (95% CI: 0.653–0.992) in Shikoku island, and 0.802 (95% CI: 0.124–0.970) in Kyushu island. The lack of island-level data did not allow us to test whether limited changes in susceptibility estimated for the Ryukyu island are correlated with a lack of increase in childcare facilities in Ryukyu island; however, counterfactual simulations show that an increase in susceptibility would have been able to shift the RSV outbreaks to spring in Ryukyu island, consistent with predictions for other islands based on the quadratic humidity-transmission relationship (Supplementary Figure S6B). Counterfactual simulations show that an increase in susceptibility would not be able to cause a sudden shift in the RSV outbreak seasonality in Hokkaido island, illustrating that the semiannual pattern in seasonal transmission is necessary to cause a sudden shift in outbreak seasonality.

## Discussion

We present an epidemiological analysis of RSV outbreaks in Japan combining spatiotemporal observations with dynamical disease modeling. Our analysis revealed semiannual cycles in seasonal RSV transmission in four major islands (Honshu, Shikoku, Kyushu, and Ryukyu), which correlate with specific humidity and temperature. We found that these semiannual cycles allowed a sudden shift in the seasonality of RSV outbreaks in response to an increase in RSV transmission—this increase could be captured by either through changes in susceptibility or transmissibility. We hypothesize that an increase in childcare capacity through Comprehensive

309 Support System for Children and Childcare may have contributed to the increase in  
310 RSV transmission [? ].

311 Our analysis revealed considerable heterogeneity in epidemic dynamics across  
312 major islands in Japan. We showed that these differences could be explained by  
313 the differences in underlying seasonal transmission. Notably, we found a robust,  
314 quadratic relationship between the estimated transmission rates against mean spe-  
315 cific humidity and mean temperature across four major islands (Honshu, Shikoku,  
316 Kyushu, and Ryukyu), indicating low transmission at intermediate levels of specific  
317 humidity. These findings echo earlier studies that demonstrated similar relationships  
318 for RSV [? ] and influenza [? ? ? ? ? ]. However, given strong correlation between  
319 specific humidity and temperature, we were unable to identify the primary environ-  
320 mental driver from the current study. Their relative contributions may be separated  
321 from data spanning a wider geographical region with more climate variability as well  
322 as epidemiological variability [? ].

323 The robustness of this quadratic relationship across islands exhibiting different  
324 climate conditions also suggests a possibility that climate-driven transmission may  
325 be, in part, facilitated by human behavior: for example, an increase in time spent  
326 indoors during low and high humidity seasons can contribute to increased transmis-  
327 sion. Further investigation is needed to establish the underlying mechanism behind  
328 how climate conditions affect the transmission of respiratory pathogens, especially  
329 across different environmental contexts (e.g., indoor vs outdoor). We note that this  
330 relationship is correlational, rather than causal, and therefore any other climate vari-  
331 ables (e.g., rainfall) or seasonal variation in human behavior (e.g., school terms) that  
332 correlate with seasonal variation in specific humidity and temperature will be implic-  
333 itly captured by this relationship. Understanding why the relationship between RSV  
334 transmission and humidity in Hokkaido island differs from other locations remains  
335 to be answered.

336 We hypothesized that the Comprehensive Support System for Children and Child-  
337 care may have contributed to the increase in RSV transmission, but other mecha-  
338 nisms may have also contributed. For example, an increase in inbound overseas trav-  
339 elers may have led to an increased exposure to RSV among adults thereby increasing  
340 the total amount of transmission [? ]. However, RSV typically infects young chil-  
341 dren [? ] and contact rates among young children and adolescents (5–19) are much  
342 higher than among older adults in Japan [? ], suggesting that an increase in  
343 inbound overseas travelers may have smaller effects than an increase in childcare at-  
344 tendance on overall RSV dynamics. Another competing hypothesis that could lead  
345 to an increase in transmission would be strain evolution: for example, one study  
346 noted that L172Q/S173L mutant strains of RSV B that became dominant around  
347 2016 had reduced susceptibility against monoclonal antibodies [? ]. However, it is  
348 not yet clear how this mutation translates to susceptibility against infection-derived  
349 antibodies. While our findings are consistent with those by [? ], who also pointed  
350 out the association between an increase in childcare attendance and a large change  
351 in the seasonality of Kawasaki disease in Japan, we cannot rule out the possibility

that other factors, such as increase in air travel [?] or strain evolution [?], could have contributed to the increase in overall transmission.

Our sensitivity analysis revealed that an increase in transmissibility, rather than an increase in susceptibility, can also explain a sudden shift in seasonality in RSV outbreaks. In fact, both models provide evidence for increased contact rates. Specifically, the total rate of RSV transmission can be written as  $\hat{\beta}cSI/N$ , where  $\hat{\beta}$  represents the rate of transmission per contact,  $c$  represents the contact rate,  $S$  represents the number of susceptible individuals,  $I$  represents the number of infected individuals, and  $N$  represents the total population size. Then, an increase in contact rate  $c$  can be captured by either increase in  $S$  (the original model) and  $\hat{\beta}$  (the sensitivity analysis). Therefore, conclusions from both models are consistent with our hypothesis that an increase in contact rates, especially among children, contributed to the sudden change in RSV seasonality. Detailed age structured surveillance data, alongside age structured model, will be needed to properly assess the relative contribution of potential mechanisms, such as increase in childhood mixing patterns or increase in inbound overseas travelers.

Our model-based estimate of the initial susceptible fraction range between 8%–11%. However this estimate must be interpreted with care as it represents effective changes in population-level susceptibility, which can depend on many factors such as immunological and behavioral changes, rather than an actual susceptible fraction. Therefore, these estimates are not directly comparable to serological estimates of proportion seronegative. For example, [?] recently estimated that ~90% of individuals above 3 years of age have been infected at least once by analyzing anti-pre-F protein antibody concentration data from the Netherlands [? ?], but individuals with RSV antibodies can still permit reinfection. A cross-section serological assay from healthcare and non-healthcare workers at a pediatric medical facility in Japan found that ~95% of the participants were seropositive but only 63.5% had greater than 8 fold neutralizing antibody titers [?], suggesting challenges in translating serological data to susceptible fraction. While our estimates are not inconsistent with these data, detailed age structured serological data and model are needed to accurately estimate changes in population-level susceptibility against RSV.

Interventions to slow the transmission of COVID-19 have disrupted the circulation of many pathogens [? ? ? ?], including RSV epidemics in Japan. This disruption has added major challenges to predicting future outbreaks, which prevented us from making long-term predictions. Continued analysis of RSV dynamics in the post-COVID period, particularly with regard to whether RSV outbreaks in Japan return to fall or winter outbreaks, may help further validate our models.

Our analysis relied on several simplifying assumptions. For example, our model assumed that the waning of immunity can render previously infected individual to become fully susceptible to reinfections; this approximation allowed us to reconstruct the dynamics of susceptible hosts more easily. In practice, immunity is likely more complex with secondary infections being less susceptible and transmissible than primary infections [?]. Other studies have also suggested the importance of interaction

between RSV A and B [? ? ] as well as competition between RSV and human metapneumovirus [? ]; our model did not account for such strain dynamics. We also did not account for explicit spatial structure or underlying stochasticity of the system. Therefore, our estimates of transmission rates must be interpreted with caution as they may implicitly capture factors that we did not account for explicitly, including strain dynamics, spatial structure, and exogenous transmission. Our analyses also primarily focused on three major islands (Honshu, Shikoku, and Kyushu), necessitating a better understanding of RSV dynamics in Hokkaido and Ryukyu islands; however, we note that these three islands make up  $> 95\%$  of the population in Japan, meaning that our analyses capture the majority of RSV transmission in Japan. Despite these limitations, our model likely represents a parsimonious approximation of the complex host-pathogen interactions, allowing us to draw general conclusions about how interactions between endogenous (contact rates) and exogenous (climate-driven factors) factors can give rise to a sudden dynamical transition.

Understanding how endogenous and exogenous factors shape epidemic dynamics is critical to predicting future outbreaks and making public health decisions. Our analysis shows that the interplay between climate-driven transmission and subtle changes in contact rates can cause a sudden transition in epidemic dynamics. More broadly, our analysis demonstrates that detailed epidemiological time series data can allow us to tease apart endogenous and exogenous factors in explaining dynamical transitions, offering unique insights into a long-standing ecological question.

## Methods

### Epidemiological data

The Japan prefecture-level weekly time series of RSV cases comes from the National Institute of Infectious Diseases (NIID). The NIID issues Infectious Diseases Weekly Report (IDWR) every week, which includes sentinel-reporting diseases. Specifically, RSV infections are reported through  $\approx 3000$  pediatric sentinel sites, which cover around 10% of pediatric institutions in Japan [? ]. We downloaded all available IDWR surveillance tables for sentinel-reporting diseases from the beginning of 2013 to end of 2023 from <https://www.niid.go.jp/niid/en/surveillance-data-table-english.html> and extracted RSV time series from these tables.

### Demographic data

Population sizes for each prefecture as of 2022 were obtained from Statistics Bureau of Japan website (<https://www.e-stat.go.jp/en>). Statistics on the number of children attending childcare facilities were obtained from the Children and Families Agency website ([cfa.jp](https://cfa.jp)).

## Climate data

Both specific humidity and surface air temperature data used in this study are from European Centre for Medium Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) [? ]. The original data are hourly with a horizontal resolution of about 31km. We first resample the hourly data to obtain daily mean values and then perform spatial average over cell grids within each prefecture in Japan. We further summarized the daily time series into weekly mean values in each prefecture, which were further averaged over to obtain weekly mean values in each island.

## Center of gravity and epidemic trough

In order to characterize changes in the timing of the epidemic, we quantified the center of gravity of RSV cases for each RSV season at each prefecture. Here, we excluded the Okinawa prefecture, which is the southernmost prefecture of Japan, due to differences in RSV seasonality: in contrast to all other prefectures that exhibit winter outbreaks, summer outbreaks are observed in the Okinawa prefecture. To compute the center of gravity, we defined the RSV season from week 27 of the current year to week 26 of the next year and numbered each week of season from 1 (starting from week 27 of a given year) to 52 (ending at week 26 of the following year); this simplification allows us to track changes in RSV seasonality in a consistent manner. Then, for each season, center of gravity was calculated by taking the weighted mean of the week of season:

$$\text{center of gravity} = \frac{\sum_{w=1}^{52} w \times \text{cases during week } w}{\sum_{w=1}^{52} \text{cases during week } w} \quad (8)$$

where  $w$  represents the week of season, ranging from 1 (starting from week 27 of a given year) to 52 (ending at week 26 of the following year). We added 26 to the resulting center of gravity to convert the estimates to be in the units of regular weeks (rather than the week of season). For each season, we also quantified the corresponding epidemic trough, which represents the minimum value of weekly cases. For the 2019–2020 season, we took the minimum cases before 2020 to exclude the impact of COVID-19 interventions.

## Transmission and observation model

To model the population-level spread of RSV in Japan, we extended the standard Susceptible-Infected-Recovered-Susceptible (SIRS) model to account for non-sinusoidal seasonal transmission rates and changes in transmission patterns due to COVID-19 intervention measures. Specifically, the discrete-time SIRS model is given

463 by:

$$\text{FOI}(t) = \frac{\beta(t)(I(t) + \omega)}{N} \quad (9)$$

$$\Delta S(t) = [1 - \exp(-(\text{FOI}(t) + \mu)\Delta t)] S(t - \Delta t) \quad (10)$$

$$N_{SI}(t) = \frac{\text{FOI}(t)\Delta S(t)}{\text{FOI}(t) + \mu} \quad (11)$$

$$\Delta I(t) = [1 - \exp(-(\gamma + \mu)\Delta t)] I(t - \Delta t) \quad (12)$$

$$N_{IR}(t) = \frac{\gamma\Delta I(t)}{\gamma + \mu} \quad (13)$$

$$\Delta R(t) = [1 - \exp(-(\nu + \mu)\Delta t)] R(t - \Delta t) \quad (14)$$

$$N_{RS}(t) = \frac{\nu\Delta R(t)}{\nu + \mu} \quad (15)$$

$$S(t) = S(t - \Delta t) + \mu N - \Delta S(t) + N_{RS}(t) \quad (16)$$

$$I(t) = I(t - \Delta t) - \Delta I(t) + N_{SI}(t) \quad (17)$$

$$R(t) = R(t - \Delta t) - \Delta R(t) + N_{IR}(t) \quad (18)$$

$$(19)$$

464 Here,  $S$ ,  $I$ , and  $R$  represent the number of individuals who are susceptible, in-  
 465 fected, and recovered;  $N$  represents the total population size;  $\text{FOI}(t)$  represents the  
 466 force of infection at time  $t$ ;  $\Delta X(t)$  represents number of individuals who leave the  
 467 compartment  $X$  at time  $t$ ;  $N_{XY}(t)$  represents the number of individuals who move  
 468 from compartment  $X$  to compartment  $Y$  at time  $t$ ;  $\beta(t)$  represents the time-varying  
 469 transmission rate;  $\omega$  represents the number of imported infections;  $\gamma$  represents the  
 470 recovery rate;  $\nu$  represents the immune waning rate; and  $\mu$  represents the birth and  
 471 death rates. This model assumes a simple demography and extreme waning, which  
 472 allows immune individuals to become fully susceptible. Therefore, model paramete-  
 473 rs must be interpreted with care, especially the duration of immunity. In practice,  
 474 re-infection can still occur even under partial immunity, in which case the duration  
 475 of immunity can be shorter [? ].

476 Typically, the transmission rate is assumed to follow a sinusoidal function for  
 477 modeling endemic diseases. Instead, we decomposed  $\beta(t)$  into a product of two  
 478 separate terms:

$$\beta(t) = \beta_{\text{seas}}(t)\delta(t), \quad (20)$$

479 where  $\beta_{\text{seas}}(t)$  represents the seasonal transmission rate and  $\delta(t)$  represents relative  
 480 changes in transmission due to COVID-19 intervention measures, such that  $\delta < 1$   
 481 represents transmission reduction. A similar decomposition was recently used for  
 482 modeling the spread of *Mycoplasma pneumoniae* infections [? ].

483 First, we modeled the seasonal transmission rate  $\beta_{\text{seas}}(t)$  as a periodic function  
 484 with a period of 52 weeks ( $\beta_{\text{seas}}(t) = \beta_{\text{seas}}(t - 52)$ ) and tried to estimate a separate  
 485 value for each week. To constrain the shape of  $\beta_{\text{seas}}(t)$ , we imposed cyclic random-

486 walk priors:

$$\beta_{\text{seas}}(t+1) \sim \text{Normal}(\beta_{\text{seas}}(t), \sigma_{\text{seas}}), \quad t = 1, \dots, 51 \quad (21)$$

$$\beta_{\text{seas}}(1) \sim \text{Normal}(\beta_{\text{seas}}(52), \sigma_{\text{seas}}) \quad (22)$$

487 where the standard deviation  $\sigma_{\text{seas}}$  determines the smoothness of the seasonal trans-  
488 mission rate. We imposed a weakly informative prior on  $\sigma_{\text{seas}}$ :

$$\sigma_{\text{seas}} \sim \text{Half-Normal}(0, 1). \quad (23)$$

489 To further constrain the range of seasonal transmission rate  $\beta_{\text{seas}}(t)$ , we imposed  
490 additional priors:

$$\beta_{\text{seas}}(t) \sim \text{Normal}(8, 2). \quad (24)$$

491 Second, we assumed  $\delta(t) = 1$  for  $t < 2020$  (before the COVID-19 pandemic) and  
492 tried to estimate a separate value for  $\delta(t)$  at each week. To constrain the shape of  
493  $\delta(t)$ , we imposed random-walk priors:

$$\delta(t+1) \sim \text{Normal}(\delta(t), \sigma_{\delta}), \quad t \geq 2020 \quad (25)$$

494 where the standard deviation  $\sigma_{\text{seas}}$  determines the smoothness of the estimated  $\delta(t)$ .  
495 We imposed a weakly informative prior on  $\sigma_{\delta}$ :

$$\sigma_{\delta} \sim \text{Half-Normal}(0, 0.1). \quad (26)$$

496 To further constrain the range of estimated intervention effects  $\delta$ , we imposed addi-  
497 tional priors:

$$\delta(t) \sim \text{Normal}(1, 0.2). \quad (27)$$

498 For the analysis of Hokkaido island, we estimated  $\delta(t)$  beginning from 2017 instead  
499 of 2020 to capture the sudden transition to irregular epidemic dynamics occurred in  
500 2017; we tried fitting a model that estimated  $\delta(t)$  beginning from 2020 but found that  
501 it was unable to explain the sudden transition. For the analysis of Ryukyu island,  
502 we estimated  $\delta(t)$  beginning from 2019 instead of 2020 to capture the unusually large  
503 RSV outbreak that happened in 2019.

504 We assumed that the recovery rate  $\gamma = 1/\text{week}$  and birth/death rates  $\mu = 1/(80 \times$   
505  $52) \text{ weeks}$  are known. We imposed weakly informative priors on all other parameters:

$$1/\nu \sim \text{Half-Normal}(0, 200) \quad (28)$$

$$\omega \sim \text{Half-Normal}(0, 200) \quad (29)$$

$$s(0) \sim \text{Normal}(0.08, 0.02) \quad (30)$$

$$i(0) \sim \text{Half-Normal}(0, 0.001) \quad (31)$$

506 where  $s(0)$  and  $i(0)$  represent the initial proportion of susceptible and infected indi-  
507 viduals such that the initial conditions are given by:  $S(0) = Ns(0)$  and  $I(0) = Ni(0)$ .  
508 The initial number of recovered individuals was modeled as  $R(0) = N - S(0) - I(0)$ .

Finally, the model was fitted to case data in each island by assuming a negative binomial observation error:

$$C(t) = N_{SI}(t) \quad (32)$$

$$\text{cases}_t \sim \text{Negative-Binomial}(\rho C(t), \phi) \quad (33)$$

$$\rho \sim \text{Half-Normal}(0, 0.02) \quad (34)$$

$$\phi \sim \text{Half-Normal}(0, 10) \quad (35)$$

where  $C(t)$  represents the incidence of infection,  $\rho$  represents the under-reporting rate, and  $\phi$  represents the overdispersion parameter.

Parameter estimation was performed in a Bayesian framework using the `rstan` package version 2.32.6 [? ]. The following parameters and initial conditions were estimated simultaneously from the model: initial proportion of susceptible individuals  $s(0)$ , initial proportion of infected individuals  $i(0)$ , under-reporting rate  $\rho$ , transmission rate  $\beta$ , standard deviation in seasonal transmission rates  $\sigma_{\text{seas}}$ , overdispersion parameter  $\phi$ , duration of immunity  $1/\nu$ , number of imported infections  $\omega$ , relative changes in transmission due to COVID-19 intervention measures  $\delta(t)$ , and standard deviation in transmission changes  $\sigma_\delta$ . Convergence was assessed by ensuring low R-hat, high effective sample size, no divergent transitions, and no iterations that exceeded the maximum tree depth. The model struggled to converge for the analysis of Shikoku island—in this case, removing the  $\text{Normal}(1, 0.2)$  prior on  $\delta(t)$  allowed us to fit the model.

As a sensitivity analysis, we also considered a model that allows for changes in transmission rate before and after the sudden shift in seasonality; this analysis was performed only for Honshu, Kyushu, and Shikoku islands because the seasonal shift in RSV outbreaks was not observed in Hokkaido and Ryukyu islands. In this case, the model structure remained the same, and we tried to estimate separate  $\beta_{\text{seas}}(t)$  for two periods: winter outbreak periods (before the 26th week of 2016) and fall outbreak periods (after the 26th week of 2016). For the analysis of Shikoku island, winter and fall outbreak periods were separated at the 26th week of 2017 instead because the seasonal shift was not observed until 2017 fall.

## Simulations

We run a series of simulations to understand how the interplay between climate-driven transmission and population-level susceptibility can drive a sudden shift in the timing of an epidemic. First, we simulated the model for a year (from week 26 of the starting year to week 25 of the following year) by varying the initial conditions and computing the center of gravity. Specifically, we varied  $i(0)$  between  $1 \times 10^{-4}$  and  $3 \times 10^{-3}$  and  $s(0)$  between 0.078 and 0.105. All other parameters were set to posterior median estimates.

To further understand how the shape of seasonal transmission term affects the degree of shift in seasonality, we varied the shape of seasonal transmission term by



interpolating the estimated  $\beta_{\text{seas}}$  from Honshu and Kyshu islands, which are the two most populated islands. To do so, we first took posterior median estimates of  $\beta_{\text{seas}}$  from two islands and fitted generalized additive model with cyclic cubic spline bases to obtain smoothed estimates of  $\beta_{\text{seas}}$  for each island, which we denote as  $\beta_H$  and  $\beta_K$ , respectively. Then, we normalized seasonal transmission rates such that it has a mean of zero and has an amplitude of 1:

$$\zeta_X(t) = \frac{1}{\alpha_X} \left( \frac{\beta_X(t)}{\bar{\beta}_X} - 1 \right) \quad (36)$$

where  $\zeta_X(t)$  represents the normalized seasonal transmission pattern in island  $X$ , and  $\alpha_X = (\max(\beta_X(t)) - \min(\beta_X(t)))/2$  represents the amplitude of seasonal transmission pattern in island  $X$ . This allowed us to interpolate between two normalized seasonal terms and obtain a flexible shape for seasonal transmission rate:

$$\zeta_{\text{new}}(t) = \zeta_H(t)(1 - \theta) + \zeta_K(t)(\theta), \quad (37)$$

$$\beta_{\text{new}} = \left( \frac{\alpha_{\text{new}} \zeta_{\text{new}}(t)}{(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2} + 1 \right) \bar{\beta}_{\text{new}}, \quad (38)$$

where  $\theta$  represents the interpolation coefficient, such that  $\theta = 0$  and  $\theta = 1$  causes  $\beta_{\text{new}}$  to have the same shape as  $\beta_H$  and  $\beta_K$ , respectively and  $0 < \theta < 1$  allows us to model counterfactual transmission scenarios that interpolates between two islands. Note that  $\zeta_{\text{new}}(t)$  does not necessarily have an amplitude 1 so we divide it by  $(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2$  to ensure the amplitude of 1.

For a given value of the interpolation coefficient  $\theta$  and seasonal amplitude  $\alpha_{\text{new}}$ , we simulated two outbreaks for a year assuming  $s(0) = 0.078$  and  $s(0) = 0.105$  and computed the difference in the timing of epidemic peak. In doing so, all other parameters, including the mean transmission rate  $\beta_{\text{new}}$ , were fixed to posterior median estimates for the Honshu island.

## Regression

We performed univariate quadratic regressions for the estimated transmission rates against mean specific humidity and mean temperature, separately, and computed R squared values for each regression. We also perform bivariate quadratic regressions for the estimated transmission rates using both mean specific humidity and mean temperature as covariates. The significance of regression coefficients was determined using two-sided t-tests at the 0.05 significance level.

We performed a linear regression between the estimated proportion of susceptibles at the beginning of each season (week 26) against the number of children attending childcare facilities. For simplicity, the regression was performed using median estimates for the susceptible proportions. We did not have data on the number of children attending childcare facilities broken down by island level and so we used the national-level data instead.

## 577 Data availability

578 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parksw3/perturbation)  
579 [parksw3/perturbation](https://github.com/parksw3/perturbation)) [? ].

## 580 Code availability

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## 666 Acknowledgements

667 We acknowledge the efforts of the National Institute of Infectious Diseases, Statis-  
668 tics Bureau of Japan, and Children and Families Agency for collecting/maintaining  
669 the data used in this study and making them publicly available. E.H., B.G., and  
670 C.J.E.M. have been funded in whole or in part with Federal funds from the Na-  
671 tional Cancer Institute, National Institutes of Health, under Prime Contract No.  
672 75N91019D00024, Task Order No. 75N91023F00016. The content of this publica-  
673 tion does not necessarily reflect the views or policies of the National Institutes of  
674 Health or the Department of Health and Human Services, nor does mention of trade  
675 names, commercial products or organizations imply endorsement by the U.S. Govern-  
676 ment. S.W.P acknowledges support from Peter and Carmen Lucia Buck Foundation  
677 Awardee of the Life Sciences Research Foundation and the New Faculty Startup Fund  
678 from Seoul National University. I.H. received postdoctoral funding from the High  
679 Meadows Environmental Institute of Princeton University. B.T.G. and C.J.E.M.  
680 acknowledge support from Princeton Catalysis Initiative and Princeton Precision  
681 Health. S.C. is supported by Federal funds from the National Institute of Allergy  
682 and Infectious Diseases, National Institutes of Health, Department of Health and  
683 Human Services under CEIRR contract 75N93021C00015 - Subcontract 77789. The  
684 content is solely the responsibility of the authors and does not necessarily represent  
685 the official views of the NIAID or the National Institutes of Health.

## 686 Contributions

687 S.W.P., I.H., and B.T.G. conceived of the study. S.W.P. performed the analysis and  
688 wrote the initial draft. All authors (S.W.P., I.H., E.H., W.Y., R.E.B., G.A.V., S.C.,  
689 C.J.E.M., and B.T.G.) reviewed and edited the manuscript.

## 690 Competing Interests

691 The authors declare no competing interests.

## 692 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ( $n=52$ ). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ( $n=52$ ). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ( $n=8000$  posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ( $n=8000$  posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected  $i(0)$  and susceptible  $S(0)$  at the beginning of season on center of gravity. Points represent the estimated values for  $i(0)$  and  $s(0)$  between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the  $i(0)$  value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of  $i(0)$ . (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from  $n=8000$  posterior samples. Shaded regions represent the 95% credible interval in our estimates ( $n=8000$  posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from  $n=8000$  posterior samples. Error bars represent the 95% credible interval in our estimates ( $n=8000$  posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.